Sales & Customer Insight Analysis

1. Project Overview

This project analyzes coffee shop sales data to derive insights into business performance and customer behavior using transactional data from 20,000 purchases across various product categories. The goal is to uncover insights into sales distribution across product categories and cities, along with customer demographics, payment preferences, and the impact of external factors like weather conditions to guide strategic business decisions.

2. Dataset Summary

-Rows: 20,000
- Columns: 18
- Key Features:

- Customer demographics (Age, Gender, Location, loyalty members)
- Purchase details (Item Purchased, Category, Purchase Amount)
- Shopping behavior (Discount Applied, Payment Method, No. of Purchases, Weather conditions)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Google Colab:

- Data Loading: Imported the dataset using pandas.

store_id	city	country	store_type	product_category	product_name	unit_price	quantity	discount_applied	payme
35	Melbourne	AUS	Mall Kiosk	Coffee	Double Espresso	3.04	1	True	Credit Card
25	Manchester	UK	Standalone	Coffee	Single Espresso	2.38	2	False	Debit Card
23	Manchester	UK	Standalone	Tea	Large Matcha Latte	4.26	2	False	Mobile Wallet
38	Toronto	CAN	Standalone	Coffee	Medium Cappuccino	3.68	2	False	Credit Card
9	Los Angeles	USA	Mall Kiosk	Coffee	Small Cappuccino	3.42	1	False	Mobile Wallet

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

	transaction_id	timestamp	store_id	city	country	store_type	product_category	product_name	unit_price	quantity	discou
count	20000.000000	20000	20000.000000	20000	20000	20000	20000	20000	20000.000000	20000.00000	
unique	NaN	19992	NaN	9	4	3	6	43	NaN	NaN	
top	NaN	2023-08-12 09:23:50	NaN	New York	USA	Standalone	Coffee	Croissant	NaN	NaN	
freq	NaN	2	NaN	2288	6685	12347	9216	664	NaN	NaN	
mean	20000.500000	NaN	23.072950	NaN	NaN	NaN	NaN	NaN	4.049521	1.71265	
std	5773.647028	NaN	13.097332	NaN	NaN	NaN	NaN	NaN	1.982984	1.10808	
min	10001.000000	NaN	1.000000	NaN	NaN	NaN	NaN	NaN	1.640000	1.00000	
25%	15000.750000	NaN	12.000000	NaN	NaN	NaN	NaN	NaN	2.820000	1.00000	
50%	20000.500000	NaN	23.000000	NaN	NaN	NaN	NaN	NaN	3.690000	1.00000	

● Column Standardization: Renamed columns to snake case for better readability and documentation.

```
Index(['transaction_id', 'timestamp', 'store_id', 'city', 'country',
       'store_type', 'product_category', 'product_name', 'unit_price',
       'quantity', 'discount_applied', 'payment_method', 'customer_id',
       'customer_age_group', 'customer_gender', 'loyalty_member',
       'weather_condition', 'temperature_c', 'holiday_name', 'total_amount'],
      dtype='object')
```

● Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

```
import sqlite3

# Create connection
conn = sqlite3.connect('sales.db')

# Save dataframe into SQL table
df.to_sql('sales', conn, if_exists='replace', index=False)

print("✅ SQL table created!")
```

```
✅ SQL table created!
```

4. Data Analysis using SQL

We performed structured analysis in Google Colab to answer questions:

1. Avg Revenue acc to Weather – Compared average revenue generated in different weather conditions.

	weather_condition	avg_sales
0	None	7.091534
1	Cloudy	6.889769
2	Rainy	6.670892
3	Snowy	7.222351
4	Sunny	6.985698

2. Total Revenue made by Product category – Identified the total sales done by each product .

	product_category	total_sales
0	Coffee	58037.99
1	Tea	29622.10
2	Sandwich	17589.90
3	Merchandise	12218.84
4	Pastry	10049.26
5	Smoothie	9491.18

3. Products bought by No. of customers – no. of customers who bought a single product .

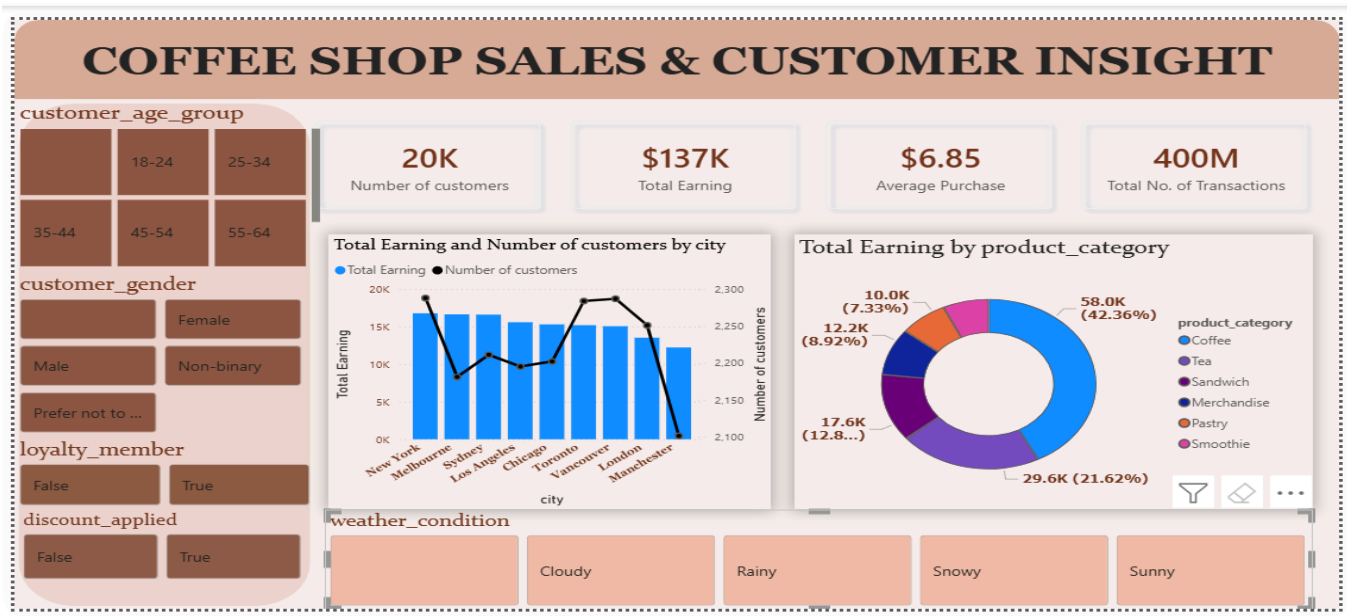
	product_name	num_customers
0	Croissant	659
1	Medium Cappuccino	649
2	Double Espresso	649
3	Small Mocha	635
4	Blueberry Muffin	635
5	Small Latte	634
6	Scone	634
7	Medium Americano	633
8	Medium Latte	632
9	Large Cappuccino	629
10	Single Espresso	621

4. Customers with Loyalty Memberships – Transactions made by the customers with or without membership there total and average spends.

	loyalty_member	transactions	total_spent	avg_spent
0	0	14139	98372.27	6.957513
1	1	5861	38637.00	6.592220

5. Total sales in each city: total sales made by customers in each city.

	city	total_sales
0	New York	16789.12
1	Melbourne	16650.31
2	Sydney	16612.94
3	Los Angeles	15596.14
4	Chicago	15309.60
5	Toronto	15209.40
6	Vancouver	15068.80
7	London	13538.02
8	Manchester	12234.94

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.